Cambridge O Level Computer Science

Chapter 9: Databases

Lecture notes with Check-Yourself questions and a full answer key.

In this chapter you will learn about:

- The design and use of single-table databases: fields, records and validation.
- The basic data types: text/alphanumeric, character, Boolean, integer, real and date/time.
- The identification, purpose and use of primary keys.
- SQL scripts used to query data stored in a single-table database.

A database is a structured collection of data (text, numbers, pictures, anything storable) that lets people extract information to meet their needs. Relational databases with many tables are studied at A Level; for IGCSE you study single-table databases, containing only one table. Practical work with a database management system (the book uses Microsoft Access 365) is strongly recommended.

9.1 Databases

9.1.1 Single-table databases

Why are databases useful? They prevent problems because:

- Any change or addition only has to be made once, so data stays consistent.
- The same data is used by everyone.
- Data is stored only once (in relational databases), so there is no duplication.

What are databases used for? Storing information about people (hospital patients, school pupils), things (cars for sale, library books) and events (hotel bookings, race results).

Fields and records

Inside a database, data is stored in tables. Each record is a row describing one item, person or event; each field is a column holding one specific piece of information about that item. The number of fields is fixed (every record has the same fields), while the number of records varies as records are added and deleted. Tables and

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fields get meaningful names; field names are single words with no spaces, for example BedNumber.

Example: a PATIENT table with fields FirstName, FamilyName, DateOfAdmission, Consultant, WardNumber, BedNumber. A BOOK table might have Title, Author, ISBN. An APPOINTMENT table might record 'Winnie Sing (a hospital patient)', 'IGCSE Computer Science (a book)' or '15:45 on January 2020 (an appointment)' as single records.

Check yourself

1. 1. What is a database, and what is a single-table database?
2. 2. Give three reasons why databases are useful.
3. 3. Give one example each of a database storing information about people, things and events.
4. 4. What is the difference between a record and a field?
5. 5. A table has 6 fields and 8 records. How many data items does it store? If 3 records are deleted, what changes?
6. 6. Suggest suitable field names for a table storing information about cars for sale.

Validation in databases

Validation (see Chapter 7, Section 7.5) also applies to databases. Some checks are automatic in the database management software: for example, the DateOfAdmission field is automatically checked to be a valid date before it can be stored. Other checks must be set up by the database developer: for example, a validation rule allowing only whole numbers 1 to 10 in WardNumber, so an out-of-range ward number is rejected with an error message.

Check yourself

7. 1. Give one example of automatic validation and one of developer-set validation in the PATIENT table.
8. 2. A BedNumber field must hold whole numbers from 1 to 8. What validation rule would you set?
9. 3. Why must validation be set up before the database is used?

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9.1.2 Basic data types

Each field needs a data type, classifying how the data is stored, displayed and what operations it allows. An integer field is stored as a whole number and can be used in calculations. The six syllabus data types (Access names differ):

Syllabus data type	Description	Access data type
text/alphanumeric	A number of characters	short text / long text
character	A single character	short text with field size of one
Boolean	One of two values: True/False, 1/0, Yes/No	Yes/No
integer	Whole number	number formatted as fixed with zero decimal places
real	A decimal number	number formatted as decimal
date/time	Date and/or time	Date/Time

Check yourself

10. 1. Choose a suitable data type for each PATIENT field: FirstName, FamilyName, DateOfAdmission, Consultant, WardNumber, BedNumber. Give reasons.
11. 2. What is the difference between the text/alphanumeric and character data types?
12. 3. Give an example of a field where the Boolean data type would be suitable.
13. 4. Why would WardNumber be an integer rather than text?

9.1.3 Primary keys

Each record describes one item, so each item must be uniquely identifiable. A primary key is a field whose values are never repeated in the table, uniquely identifying each record. It can be an existing field if it is already unique (for example, ISBN in a book table). The PATIENT table needs an extra field, since names and other fields could repeat: a primary key field called HospitalNumber holding 8 characters starting with HN followed by 6 digits, for example HN123456.

Worked example: Which of these PATIENT fields could be a primary key: FirstName, FamilyName, DateOfAdmission, WardNumber, HospitalNumber?

FirstName: no, two patients could share a first name.

FamilyName: no, two patients could share a family name.

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DateOfAdmission: no, two patients could be admitted on the same date (for example, Farouk Khan and Ahmad Teo were both admitted on 22/11/2022).

WardNumber: no, many patients share a ward.

HospitalNumber: yes, every patient gets a unique number, so no value is ever repeated.

Check yourself

14. 1. What is a primary key, and what rule must its values follow?
15. 2. Explain why the PATIENT table needs an extra field for its primary key.
16. 3. Give an example of an existing field that could serve as a primary key, and explain why it works.
17. 4. A school table has fields FamilyName, OtherNames, StudentID, DateOfBirth, CurrentClass. Which would you choose as the primary key? Give a reason.

9.1.4 SQL

Structured Query Language (SQL), pronounced 'es-queue-el', is the standard query language for obtaining useful information from a database. An SQL script is a list of SQL commands performing a given task, often stored in a file so it can be reused. You can write scripts in Access, MySQL or SQLite, but commands differ slightly between applications, so check what your software supports. You must be able to understand SQL scripts and identify their output.

The SQL statements you need:

Statement	Description
SELECT	Fetches specified fields (columns) from a table; queries always begin with SELECT.
FROM	Identifies the table to use.
WHERE	Includes only records (rows) matching a given condition.
ORDER BY	Sorts the results by a given column, alphabetically or numerically.
SUM	Returns the sum of all values in a field (column); used with SELECT.
COUNT	Counts the records (rows) where the field matches a specified condition; used with SELECT.

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